PRE-MOCK EXAMINATIONS 2026

UGANDA ADVANCED CERTIFICATE OF EDUCATION

CHEMISTRY (Theory)
Paper 1 — P525/1

Time: 2 hours 45 minutes

NAME …… SIGN……

INSTRUCTIONS TO CANDIDATES

- This examination paper consists of two sections, A and B. It has six examination items.
- Section A has two compulsory items.
- Section B has two parts, I and II. Respond to only one item from each part.
- Respond to four items in all.
- Any additional item(s) responded to will not be scored.
- Begin each item on a fresh page.

SECTION A

Respond to both items from this section.

Item 1.

The increasing cost of conventional fuels has encouraged many institutions in Uganda to explore alternative sources of energy. A boarding school installed a biogas digester that uses animal waste and food remains to produce methane for cooking.

During operation of the digester, the following observations were made:

i. The temperature of the contents of the digester increased during decomposition of the waste.
ii. Gas production was initially high after fresh waste was added but gradually decreased with time.
iii. Food cooked faster when the supply of methane to the burner was increased.

The following information about methane production and combustion is provided.

Data set 1

$C_6H_{12}O_6(s) \rightarrow 3CH_4(g) + 3CO_2(g)$

Substance	Enthalpy of formation / kJ mol^{-1}
$C_6H_{12}O_6(s)$	−1273
$CH_4(g)$	−75
$CO_2(g)$	−394

Data set 2

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$

The following experimental results were obtained for the combustion process.

Experiment	[CH_4] / mol dm^{-3}	[O_2] / mol dm^{-3}	Initial rate / mol dm^{-3} s^{-1}
A	0.10	0.10	2.0×10^{-4}
B	0.20	0.10	4.0×10^{-4}
C	0.20	0.20	1.6×10^{-3}

Task

a) Analyse the observations made during the production and use of the biogas, using the information provided.

b) Assess the conditions and practices that the school should adopt to maximise the production and efficient use of methane as a fuel.

Item 2.

A fruit-processing company produces bottled passion-fruit juice for sale in supermarkets. The company uses benzoic acid and sodium benzoate to control the acidity of the juice and sugar to improve its taste.

Customers complained that one batch of the juice tasted unusually sour and froze more readily than another batch when placed under similar refrigeration conditions.

The company investigated two formulations used to prepare **100 cm^3** samples of juice.

Substance	Sample P	Sample Q
Benzoic acid / g	1.22	1.22
Sodium benzoate / g	0.288	0.288
Sugar / g	8.0	24.0

The pKa of benzoic acid is **4.20**.

The recommended pH range for the juice is **3.5–4.0**.

The relative molecular masses are:

$C_6H_5COOH = 122$
$C_6H_5COONa = 144$

Task

a) Assess the customers' concerns about the acidity and freezing behaviour of the two juice samples.

b) Evaluate the benefits and possible consequences of the additives used in the juice, and recommend appropriate practices for their use.

SECTION B

PART I

Respond to only one item from this part.

Item 3.

A ceramic and glass manufacturing company uses compounds of silicon and other Period 3 elements in its production processes.

During a study visit, learners observed that:

i. silicon(IV) oxide was used as a major raw material in making heat-resistant products;
ii. magnesium oxide was used to neutralise some acidic industrial wastes;
iii. aluminium oxide could not be used effectively to treat strongly alkaline waste;
iv. gaseous oxides of sulphur and phosphorus produced during some processes could contaminate rainwater and nearby water sources.

The company also uses chlorine-containing substances during water treatment.

Task

a) Analyse the observations made by the learners in terms of the structure, bonding and chemical properties of the substances involved.

b) Explain, using appropriate chemical equations, the reactions relevant to the treatment of the industrial wastes and water.

c) Evaluate the environmental importance and possible dangers associated with the substances used or produced by the company, and suggest suitable mitigation measures.

Item 4.

A company plans to construct a pedestrian bridge across a river and is considering the use of **aluminium** and **iron** as the main structural materials.

During discussions, the engineers noted that:

i. the bridge would be exposed continuously to air and moisture;
ii. iron components could be protected by attaching blocks of magnesium;
iii. aluminium normally remains resistant to further corrosion despite being a reactive metal;
iv. a worker suggested replacing aluminium with beryllium because the two elements occur in related positions in the Periodic Table.

The company is also concerned about the environmental and health effects associated with extracting, processing and using these metals.

Task

a) Analyse the physical properties of aluminium and iron in relation to their structures and suitability for bridge construction.

b) Explain the chemical behaviour of the metals in the stated environment, including the methods by which corrosion may be controlled.

c) Assess the suggestion to use beryllium and evaluate the environmental and health impacts associated with the materials under consideration.

PART II

Respond to only one item from this part.

Item 5.

Plastic waste from packaging materials has increasingly accumulated around markets and drainage channels in an urban area. An environmental chemist identified two major polymers, **X** and **Y**.

Polymer X has the repeating unit:

$[–CH_2–CH_2–]_n$

Polymer Y has the repeating unit:

$[–CH(C_6H_5)–CH_2–]_n$

Further investigations showed that:

i. the monomer used to manufacture X can be prepared from ethanol;
ii. ethanol can be manufactured from ethanoic acid through suitable chemical transformations;
iii. the monomer used to manufacture Y is obtained from an aromatic compound;
iv. an aromatic ketone used in another industrial process can be prepared from benzene using an acyl chloride derived from ethanoic acid.

Task

a) Analyse the structural features and properties of polymers X and Y that contribute to their accumulation in the environment.

b) Analyse the reaction pathways involved in:

- the formation of the monomers of X and Y;
- the polymerisation of the monomers;
- the conversion of ethanoic acid into the stated organic intermediates; and
- the formation of the aromatic ketone.

Include appropriate reagents, conditions, equations and reaction mechanisms.

c) Evaluate the useful applications and the environmental and health impacts of polymers X and Y, and recommend appropriate waste-management practices.

Item 6.

A pharmaceutical company extracts active ingredients from plant materials using organic solvents. The company also manufactures several organic compounds for use as solvents and chemical intermediates.

The following compounds were found in the company's chemical store:

- **Compound A:** a straight-chain hydrocarbon containing six carbon atoms and having a relative molecular mass of 86.
- **Compound B:** an aromatic hydrocarbon containing seven carbon atoms and having a relative molecular mass of 92.
- **Compound C:** an alcohol containing two carbon atoms and having a relative molecular mass of 46.

The company uses suitable starting materials to manufacture:

- nitrobenzene;

- ethyl ethanoate;
- propanone; and
- phenylethanone.

Before selecting any compound for industrial use, the company assesses its physical properties, chemical behaviour, usefulness and possible effects on human health and the environment.

Task

a) Identify compounds A, B and C, and examine their structures and physical properties to determine their suitability as solvents for extracting organic substances from plant materials.

b) Analyse the reaction pathways and mechanisms involved in the preparation of nitrobenzene, ethyl ethanoate, propanone and phenylethanone from suitable starting materials related to the compounds in the scenario.

Include appropriate reagents, conditions, equations and mechanisms.

c) Evaluate the practical applications and the health and environmental impacts of the organic compounds involved, and recommend appropriate measures for their safe handling and use.

END